Gene-Level Engineering Strategy for Lysine Biofortification in Wild-Type Maize (GPR-Associated Targets Only)

TIER 1: HIGH-IMPACT KNOCKOUT TARGETS

(Block fatty acid biosynthesis entry - 100% biomass retained)

*	**R00216[K,p]** -> 5 genes - CRISPR multiplex KO	Pyruvate Dehydrogenase (plastid)	**FC: 16,234x**	*GRMZM2G085019 +4 isozymes*
*	**R00282[K,c]** -> 1 gene - single CRISPR KO	Acetyl-CoA Carboxylase (cytosol)	**FC: 28x**	*GRMZM2G000236*

TIER 2: OVEREXPRESSION & MIMICRY TARGETS

(~10% lysine increase - enhance carbon supply & capture)

*	**R01280[K,p]**	Chorismate Synthase	**OE**	**1.10x**	*GRMZM2G002614 (+12 isozymes)*
*	**R00883[K,c]**	Phosphofructokinase	**OE**	**1.10x**	*GRMZM2G119300 or GRMZM2G149265*
*	**R00512[K,c]**	Pyruvate Kinase	**OE**	**1.10x**	*GRMZM2G067453 (+3 isozymes)*
*	**R00667[K,m]**	TCA Cycle (mito.)	**OE**	**1.10x**	*GRMZM2G080828 or GRMZM2G119583*
*	**R00342[K,p]**	Transketolase (plastid)	**MIM**	**1.10x**	*GRMZM2G035767 (+10 isozymes)*
*	**MR00422[K,p]**	Lipid Metab. (plastid)	**MIM**	**1.10x**	*GRMZM2G002656 (+14 isozymes)*

TIER 3: SYNERGISTIC GENE COMBINATIONS

(Maximum lysine gain with 100% biomass viability - all targets have known genes)

>	**KO: R00216 + KO: R00282**	Pyr DH + ACC block	**FC: 16,234x**	*GRMZM2G085019x5 + GRMZM2G000236*
>	**MIM: MR00422 + KO: R00216**	Lipid down + Pyr DH block	**FC: 16,234x**	*GRMZM2G002656 + GRMZM2G085019x5*
>	**OE: R01280 + KO: R00216**	Chorismate up + Pyr DH block	**FC: 16,234x**	*GRMZM2G002614 + GRMZM2G085019x5*
>	**OE: R00883 + KO: R00282**	PFK up + ACC block	**FC: 28x**	*GRMZM2G119300 + GRMZM2G000236*

REFERENCE: Lysine Biosynthetic Pathway Genes

DHDPS (R02292[K,p]): GRMZM2G027835 -- Committed step of lysine biosynthesis (overexpress)

DAP Decarboxylase (R00451[K,p]): GRMZM2G020446 -- Final step producing free L-lysine